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File Operations

```
root@DESKTOP-OAP8OK0:/# mkdir ~/linux_lab
root@DESKTOP-OAP8OK0:/# ls
batch38  boot  etc   init  lib64       media  opt   root  sbin  srv  test.c  usr
bin      dev   home  lib   lost+found  mnt    proc  run   snap  sys  tmp     var
root@DESKTOP-OAP8OK0:/# cd ~/linux_lab
root@DESKTOP-OAP8OK0:~/linux_lab# touch greeting.txt
root@DESKTOP-OAP8OK0:~/linux_lab# ls
greeting.txt
root@DESKTOP-OAP8OK0:~/linux_lab# nano greeting.txt
root@DESKTOP-OAP8OK0:~/linux_lab# cat greeting.txt
"Hello Linux!"
root@DESKTOP-OAP8OK0:~/linux_lab# cp greeting.txt backup.txt
root@DESKTOP-OAP8OK0:~/linux_lab# ls
backup.txt  greeting.txt
root@DESKTOP-OAP8OK0:~/linux_lab# cat backup.txt
"Hello Linux!"
root@DESKTOP-OAP8OK0:~/linux_lab# mv backup.txt old_greeting.txt
root@DESKTOP-OAP8OK0:~/linux_lab# ls
greeting.txt  old_greeting.txt
root@DESKTOP-OAP8OK0:~/linux_lab# rm old_greeting.txt
root@DESKTOP-OAP8OK0:~/linux_lab# ls
greeting.txt
root@DESKTOP-OAP8OK0:~/linux_lab# mkdir temp_dir
root@DESKTOP-OAP8OK0:~/linux_lab# ls
greeting.txt  temp_dir
root@DESKTOP-OAP8OK0:~/linux_lab# rmdir temp_dir
root@DESKTOP-OAP8OK0:~/linux_lab# ls
greeting.txt
root@DESKTOP-OAP8OK0:~/linux_lab# ls -l greeting.txt
-rw-r--r-- 1 root root 15 Nov  8 05:11 greeting.txt
```

Figure 1: Essential file commands

APT Package Management

```
root@DESKTOP-OAP8OK0:~/linux_lab# sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
54 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@DESKTOP-OAP8OK0:~/linux_lab# sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 54
root@DESKTOP-OAP8OK0:~/linux_lab# gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

root@DESKTOP-OAP8OK0:~/linux_lab# apt search python3

alembic/resolute 1.18.4-1 all
  lightweight database migration tool for SQLAlchemy

antimony/resolute 0.9.3-2.1build2 amd64
  Computer-aided design CAD tool
```

Figure 2: APT commands

C Compilation

Step 1: Write the code.

```
root@DESKTOP-OAP8OK0:~/linux_lab# touch hello.c
root@DESKTOP-OAP8OK0:~/linux_lab# ls
greeting.txt  hello.c
root@DESKTOP-OAP8OK0:~/linux_lab# nano hello.c
root@DESKTOP-OAP8OK0:~/linux_lab# cat hello.c
#include<stdio.h>

int main() {
        int a = 10, b = 5;

        printf("Hello, Linux Lab!\n");
        printf("Sum of %d and %d is %d\n", a, b, a + b);

        return 0;
}
```

Figure 3: Create source file and edit and view contents inside hello.c file.

Step 2: Compile and run.

```
root@DESKTOP-OAP8OK0:~/linux_lab# gcc hello.c -o hello_program
root@DESKTOP-OAP8OK0:~/linux_lab# ls -l hello_program
-rwxr-xr-x 1 root root 15992 Nov  8 05:48 hello_program
root@DESKTOP-OAP8OK0:~/linux_lab# ./hello_program
Hello, Linux Lab!
Sum of 10 and 5 is 15
root@DESKTOP-OAP8OK0:~/linux_lab# gcc -Wall hello.c -o hello_program_warn
root@DESKTOP-OAP8OK0:~/linux_lab# ./hello_program_warn
Hello, Linux Lab!
Sum of 10 and 5 is 15
```

Figure 4: Compile and execute.

Step 3: Modify and recompile.

```
root@DESKTOP-OAP8OK0:~/linux_lab# nano hello.c
root@DESKTOP-OAP8OK0:~/linux_lab# cat hello.c
#include<stdio.h>

int main() {
        int a = 200, b = 5;

        printf("Hello, Linux Lab!\n");
        printf("Sum of %d and %d is %d\n", a, b, a + b);

        return 0;
}
root@DESKTOP-OAP8OK0:~/linux_lab# gcc -Wall hello.c -o hello_program
root@DESKTOP-OAP8OK0:~/linux_lab# ./hello_program
Hello, Linux Lab!
Sum of 200 and 5 is 205
```

Figure 5: Iterative development

Cleanup and verification

```
root@DESKTOP-OAP8OK0:~/linux_lab# cd ~
root@DESKTOP-OAP8OK0:~# ls -lh ~/linux_lab/
total 40K
-rw-r--r-- 1 root root  15 Nov  8 05:11 greeting.txt
-rw-r--r-- 1 root root 150 Nov  8 05:51 hello.c
-rwxr-xr-x 1 root root 16K Nov  8 05:51 hello_program
-rwxr-xr-x 1 root root 16K Nov  8 05:49 hello_program_warn
root@DESKTOP-OAP8OK0:~# cat ~/linux_lab/hello.c
#include<stdio.h>

int main() {
        int a = 200, b = 5;

        printf("Hello, Linux Lab!\n");
        printf("Sum of %d and %d is %d\n", a, b, a + b);

        return 0;
}
```

Figure 6: Final checks

Quick Reference Summary

- View: cat <file name>
- Copy: cp <filename from which you want to copy> <new filename u want to paste that copy>
- Move/Rename: mv <filename you want to rename> <new file name you want to rename it with>
- Remove directory: rmdir <directory name>
- Remove directory and its contents: rm -r <directory name>
- Install: sudo apt update && sudo apt install build-essentials
- Compile C: gcc file.c -o <executable filename>
- Check Permissions: ls -l <file name> or ls -lh ~/<directory name>/